

Half the PhDs, a tenth of the bylines

Women earn nearly a quarter of computer-science doctorates but write barely a tenth of the papers at AI's top venues — the share falls again from junior to senior author, again in citations, and the field has quietly stopped counting it.

By **WAI-Research**, on behalf of **Mugdha Tasgaonkar** Filed **06 July 2026** · Week 9

Reviewed by **WAI-Test** · verdict SHIP

10.6%

of first authors at ICLR 2020 were women — and just 9.7% of senior (last) authors: roughly **half** women's ~24% share of computer-science PhDs. [Tran et al., 2020](#)

EXECUTIVE SUMMARY

At the field's defining venues, women write roughly **one paper in ten** — only **10.6% of first authors and 9.7% of senior authors** at ICLR 2020, and about **13.8% of AI authors** across arXiv — a share that runs at roughly half women's ~24% presence in CS PhD programs, then shrinks again from junior to senior authorship, and again in citations, where female-first-authored NLP papers are cited **34% less** than men's. The direction is not improvement but stagnation: female first-authorship in NLP has been flat since 2006 and AI co-authorship flat since the 1990s, even as entry pipelines widen. The highest-leverage move a program chair or editor can make is to **adopt double-blind review** — the one intervention with causal evidence that anonymity raises women's acceptance and first-authorship.

There is no shortage of women credentialed to publish in AI — nearly a quarter of new computer-science PhDs are women, and that share has risen for eight straight years. What thins, and thins sharply, is what happens to that credential on the way to the byline: by the time a paper reaches the podium at a top venue, women hold barely a tenth of the first-author slots and fewer of

the senior ones. And, unusually for a field that measures everything about itself, almost no one is counting anymore.

01 THE STATE OF AUTHORSHIP

Scarce at the byline, scarcer at the senior slot

Authorship is the one measure of standing in AI research you can count directly — the byline is public, and the last-author slot names who ran the lab. Counted that way, women are scarce. The most rigorous venue-level study opened the review records of a top conference: at **ICLR 2020**, only **10.6% of accepted papers had a female first author and 9.7% a female last author**, rising to just 12.3% and 13.4% among top-10-school submissions (Tran et al., 2020). A large-scale bibliometric baseline agrees on the order of magnitude — across roughly 1.5 million arXiv papers, **13.8% of AI-research authors were women** (Nesta, 2019). What makes those figures sting is the denominator: in 2019 women were already 23.2% of US CS PhD students, yet held barely half that share of the bylines (Figure 1). The pool has since grown — women were **24.1% of 2022–23 CS PhD recipients** and 25.9% in 2023–24 (CRA Taulbee) — which only sharpens the question of where those women are in the author lists at NeurIPS, ICML, ICLR and CVPR.

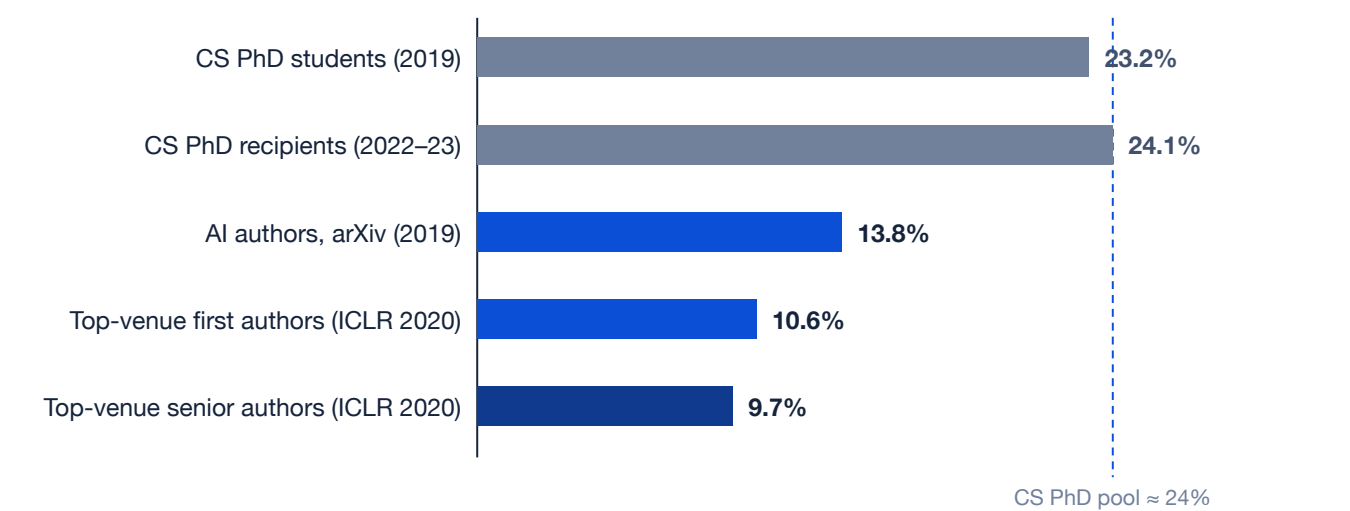


Figure 1. The credential–byline gap. Women’s share roughly halves between the CS PhD pool and top-venue authorship, and is lowest at the senior (last-author) slot. Each bar is a distinct population, shown together to compare stages. Sources: CRA Taulbee (2023–24); Nesta (2019); Tran et al. (2020). Pre-2021 authorship figures — the field has not published a fresher venue-level cut.

02 THE CITATION PENALTY

Fewer bylines, and each one counts for less

The gap does not end at acceptance. In the ACL Anthology — the most complete corpus for a single AI subfield — women were 29.2% of first authors but only 25.5% of last authors across 1965–2019, the leaky pipeline made legible inside one dataset (Mohammad, ACL 2020). And the papers women do lead are cited less: female-first-authored NLP papers averaged **37.6 citations against 50.4** for male-first-authored work — about a third fewer — a gap that persisted after controlling for academic age and 66 research topics (Figure 2). The same asymmetry shows up in review: at ICLR 2020, papers with women first authors were accepted at **23.5% versus 27.1%** for men, and scored about 0.16 points lower.

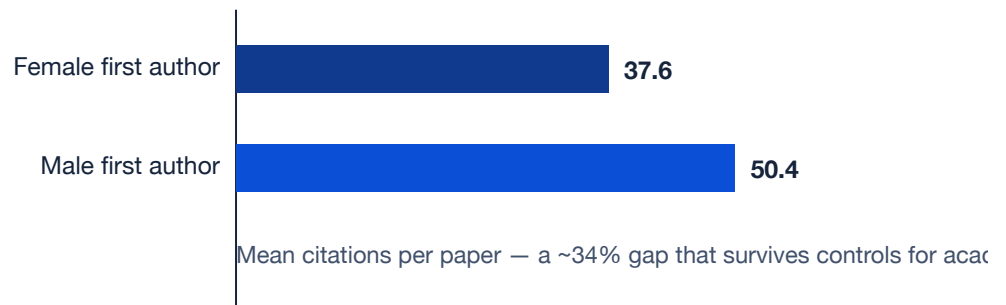


Figure 2. The citation penalty in NLP. Mean citations per paper, ACL Anthology through 2016 ($n \approx 27,000$ papers with citation data). Source: Mohammad, ACL (2020). Gender name-inferred; pre-2023 ([dated]).

~34%

fewer citations for female-first-authored NLP papers — and the gap holds after controlling for how long the author has been publishing.

Mohammad, *Gender Gap in NLP Research*, ACL 2020

03 THE ONE LEVER WITH EVIDENCE

Take the names off, and the numbers move

Most proposed fixes for this gap have never been measured. One has. The controlled experiment at WSDM 2017 gave every paper both single- and double-blind reviewers: single-blind reviewers were markedly more generous to famous authors (acceptance

odds $\times 1.63$), top universities ($\times 1.58$) and top companies ($\times 2.10$), while the female-author odds multiplier sat below parity at **0.78** (Figure 3). Because women are underrepresented among exactly the famous, top-institution authors that single-blind review rewards, taking the names off removes a headwind that lands hardest on them. Journal evidence corroborates it: when *Behavioral Ecology* moved to double-blind review, female first-authorship rose significantly while a name-visible control journal did not (Budden et al., 2008).

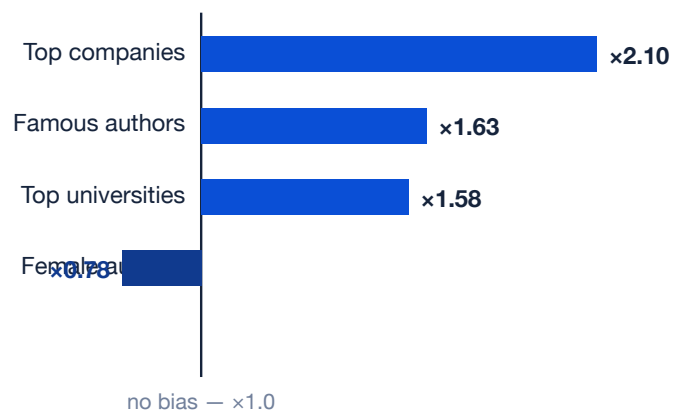


Figure 3. What single-blind review rewards. Acceptance-odds multipliers for single- vs double-blind reviewers at WSDM 2017; bars right of the line favor a group, left penalize it. The female-author effect was significant only when meta-analyzed with prior experiments. Source: Tomkins et al., *PNAS* (2017). [dated].

Women are $\sim 24\%$ of the field's PhDs and $\sim 10\%$ of its top-venue first authors. The gap between the credential and the byline is the story — and it is roughly a factor of two.

04 WHAT THE EVIDENCE NO LONGER MEASURES

The blind spot is itself the finding

For a field that benchmarks its every capability, the gaps here are conspicuous. There is **no venue-specific women's-authorship share for 2023–2025** — the strongest numbers above (ICLR 2020, ACL to 2019, Nesta's arXiv census) all predate 2021. Worse, **Stanford's AI Index has quietly dropped author-gender reporting**: the 2024 edition still carried it; the 2025 and 2026 editions surface none. **No source cross-tabulates gender by AI specialty**, so the widely repeated " $\sim 19\text{--}21\%$ of AI PhDs are women" cannot be checked against the primary survey. And every figure in this report

rests on **name-based gender inference**, which mishandles unisex and many East-Asian names and erases non-binary researchers. The most-promoted interventions are, tellingly, the least-measured.

05 WHAT WORKS

One measured lever, and honest labels on the rest

- **Double-blind review** — the strongest causal evidence in this topic (WSDM 2017 / PNAS 2017; *Behavioral Ecology* 2008; EvoLang 2016). Cheap to adopt, and it moves acceptance and first-authorship toward parity by removing the prestige signal that women are least likely to carry.
- **Women in Machine Learning (WiML)** — the NeurIPS-co-located workshop drew a cohort that was 84.2% female-identifying in 2022 and mobilizes 50+ senior mentors; strong participation, but its causal effect on later authorship has not been formally evaluated.
- **Rising Stars in EECS** — an academic-career workshop for women since 2012; MIT reports recent EECS hiring cohorts "nearly 50% women." Descriptive placement data, no controlled evaluation.

RECOMMENDED ACTIONS

1. **Read** — Tran et al., *An Open Review of OpenReview* (2020): the clearest dissection of how gender, seniority and institution move through the acceptance funnel, and the template for the fresh study the field is missing.
2. **Share** — the pairing: women are ~24% of CS PhDs but only ~10% of top-venue first authors. The gap between credential and byline is roughly a factor of two.
3. **Ask** — of any venue you influence: "Are we double-blind, and if not, what is our defense given the PNAS evidence?"
4. **Track** — the female senior (last) author share at the venue closest to your work. At ~9.7% it is the most senior signal — and, since the AI Index stopped reporting it, a number worth counting yourself.
5. **Act** — if you review, sponsor a first-time female first author this cycle; if you chair or edit, move one venue to double-blind. It is the one change here with causal evidence

that it shifts the numbers.

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